How Significant Is the Pillar 43 Sky Map? A Statistical Decomposition of the Göbekli Tepe Archaeoastronomy Claim

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Abstract

Sweatman & Tsikritsis (2017) proposed that animal carvings on Pillar 43 at Göbekli Tepe encode star constellations, dating the monument to ∼10,950 BC via precession of the equinoxes—coinciding with the proposed Younger Dryas impact event. Subsequent work by Sweatman & Gerogiorgis (2025) revised the statistical significance claim to approximately 1 in 1.4 million. We present, to our knowledge, the first independent quantitative decomposition of this claim, separating its date-coincidence and visual-similarity components and testing each under multiple null models.

Across 13 analyses, we find: (1) the date coincidence is not rare—31–67% of random constellation-cardinal assignments produce a date within ±250 years of a notable paleoclimatic event, and 43% of the 20,000–5,000 BC window lies within ±250 years of at least one such event; (2) the visual similarity between animal carvings and constellation figures is genuinely unusual ($P \approx 1$ in 391,000), but this signal is overwhelmingly carried by three of six mappings (1 in 85,184) while the remaining three contribute far less weight despite remaining statistically significant (1 in 187); (3) the reported significance varies by two orders of magnitude across defensible analyst choices. The statistical case depends primarily on three self-scored visual similarity rankings that have never been independently validated by blind evaluation. We propose that such a blind ranking experiment would be the decisive test. Code and data are publicly available.

Keywords: archaeoastronomy, Göbekli Tepe, statistical significance, look-elsewhere effect, precession, multiverse analysis

1 Introduction

Göbekli Tepe, a monumental Neolithic site in southeastern Turkey, has produced some of the most debated archaeological interpretations of the last decade. Among them, the proposal by Sweatman and Tsikritsis (2017) stands out for its specificity: the animal carvings on Pillar 43 in Enclosure D represent star constellations, and the scene encodes the date of the summer solstice at approximately 10,950 BC via precession of the equinoxes. This date coincides with the proposed Younger Dryas impact event (Kennett et al., 2015), leading the authors to interpret the pillar as a memorial to a cosmic catastrophe. The claim was published in *Mediterranean Archaeology and Archaeometry* and has since been extended to European Palaeolithic cave art (Sweatman & Coombs, 2019), a lunisolar calendar embedded in the same pillar's geometric symbols (Sweatman, 2024), and further revised probabilistic analyses (Sweatman & Gerogiorgis, 2025).

The excavation team at Göbekli Tepe responded with a qualitative rebuttal (Notroff et al., 2017), raising objections about pillar repositioning during multi-phase rebuilding, a 700–1,000 year gap between the proposed date and the site's radiocarbon chronology, and the "incredibly arbitrary" assumption that Neolithic hunters recognized Greek-derived constellations. These are serious concerns. But neither the rebuttal nor any subsequent work performed the quantitative test that would settle the debate: an independent evaluation of whether Sweatman's statistical evidence is as strong as claimed, or whether it arises readily from the combination of a dense event record, subjective visual matching, and researcher degrees of freedom.

This paper fills that gap. We decompose Sweatman's significance claim into its two constituent parts—a date-coincidence component and a visual-similarity component—and test each independently under multiple null models, sensitivity assumptions, and analyst-choice paths. We then reassemble the joint probability and examine how it varies across the space of defensible analytical decisions.

Our central finding is that the date component is readily obtained under the null, the visual component is concentrated in three of six mappings, and the full significance is highly conditional on self-scored rankings and analyst choices. The paper proceeds as follows: Section 2 provides background on the site, the claim, and the debate. Section 3 describes our methods. Section 4 presents results from 13 analyses. Section 5 discusses implications, limitations, and the blind-test proposal.

2 Background

2.1 Göbekli Tepe and Pillar 43

Göbekli Tepe is located near Şanlıurfa in southeastern Turkey, approximately 15 km northeast of the city centre. Excavations led by Klaus Schmidt (DAI) from 1995 until his death in 2014, and continued by the Şanlıurfa Museum and DAI, have uncovered a sequence of roughly circular enclosures formed by T-shaped megalithic pillars and rough stone walls (Schmidt, 2003; Dietrich et al., 2012). The site's earliest layers are radiocarbon-dated to approximately 9,530 ± 215 BCE (Dietrich et al., 2013), making it one of the oldest known monumental constructions.

Pillar 43 is embedded in the northwestern portion of Enclosure D's perimeter wall. Its western broadside carries an elaborate relief scene: a large vulture or eagle with a circular disc balanced on its wing, a scorpion below, a canid (wolf or dog) to the left, a tall bird grasping a fish or snake to the right, a partially obscured swimming bird (duck or goose) at the bottom, and several smaller figures. The pillar's narrow faces carry additional carvings including snakes and a spider. A headless human figure appears at the base. The relief extends across both broad and narrow faces, making Pillar 43 one of the most densely decorated elements at the site.

The excavation team interprets the scene in the context of Pre-Pottery Neolithic mortuary ritual and excarnation practices (Peters & Schmidt, 2004; Notroff et al., 2016). The vulture and headless man are read as references to a documented PPN funerary tradition involving defleshing by raptors. The disc is sometimes interpreted as a severed head.

2.2 Sweatman's astronomical interpretation

Sweatman and Tsikritsis (2017) proposed a radically different reading. They identified eight animal symbols on Pillar 43 as representations of star constellations, using the modern Western constellation set as rendered in the Stellarium planetarium software. The core mappings are shown in Table 1.

Table 1: Sweatman's animal-constellation mappings for Pillar 43. Ranks are from Sweatman & Gerogiorgis (2025), indicating each constellation's position in a similarity ranking against all 44 constellations visible from Göbekli Tepe's latitude (37.2°N). The vulture is excluded from statistical testing (used as the date-determining anchor). The duck/goose is excluded due to partial obscuration.

Animal	Constellation	Rank (of 44)	Role
Vulture/eagle	Sagittarius	—	Anchor (excluded)
Scorpion	Scorpius	1	Tested
Wolf/dog	Lupus	1	Tested
Tall bird + snake	Ophiuchus	1	Tested
Bending bird	Pisces	4	Tested
Ibex/quadruped	Gemini	5	Tested
Frog/bear	Virgo	6	Tested
Duck/goose	Libra	—	Excluded (obscured)

The date is derived through five steps. First, the circular disc on the vulture's wing is identified as the Sun, supported by a similar disc appearing next to a crescent (moon) on Pillar 18. Second, the vulture represents the Sagittarius constellation, specifically the "teapot" asterism. Third, the Sun positioned within Sagittarius indicates the summer solstice, since the winter solstice in Sagittarius would yield approximately 2000 AD. Fourth, due to the precession of Earth's rotational axis (period $\approx$ 25,772 years), the summer solstice sun was positioned in Sagittarius at approximately 10,950 $\pm$ 250 BC. Fifth, three "handbag" symbols in the upper panel are interpreted as the constellations marking the other three cardinal days (Pisces for spring equinox, Gemini for winter solstice, Virgo for autumn equinox), cross-validating the date.

2.3 The probability calculation and its revisions

The statistical case has been revised four times across eight years, with each revision reducing the claimed significance.

The original calculation (Sweatman & Tsikritsis, 2017) reported approximately 1 in 100 million. This was based on ranking 12 Göbekli Tepe animal symbols against 8 constellations, computing a score Z from summed ranks, and counting the fraction of $12^8 \approx 430$ million combinations scoring as well or better.

Sweatman and Coombs (2019) revised this to approximately 1 in 140 million, adding a spatial correlation factor and correcting for pool size (13 symbols instead of 12).

Sweatman and Gerogiorgis (2025) identified four flaws in the original calculation: subjective ranking without systematic justification, incorrect counting of tied combinations, circular use of the vulture both to orient the scene and to contribute to the probability estimate, and the absence of Bayesian priors. After corrections—excluding the vulture from the test, expanding the comparison pool to 44 visible constellations, and providing detailed per-constellation similarity justifications—the revised figure stands at approximately 1 in 1.4 million. This is the figure we take as our starting point.

2.4 The Notroff et al. rebuttal

The DAI excavation team raised several objections (Notroff et al., 2017). The pillars in Enclosure D are not in their original positions due to documented multi-phase rebuilding. The proposed date of 10,950 BC is 700–1,000 years older than the oldest radiocarbon date for Enclosure D. The assumption that Neolithic hunter-gatherers recognized the same constellations as ancient Greeks is described as "incredibly arbitrary." The analysis selects a few pillars from a site containing over 60 pillars with similar animal iconography.

These objections are qualitative. Sweatman and Tsikritsis (2017b) responded that probabilistic analysis should take priority over opinion in a scientific debate, and that the radiocarbon date applies to the mortar of the rough stone wall, not to the pillar carvings themselves. Neither side performed the statistical test that would resolve the disagreement: an independent evaluation of how rare the claimed coincidences actually are under well-specified null models.

3 Methods

3.1 Precession engine

We model the precession of the equinoxes as a linear shift in the ecliptic longitude of the vernal equinox at a rate of $360°/25{,}772 \approx 0.01397°$ per year. For a given year BC and cardinal day (spring equinox, summer solstice, autumn equinox, or winter solstice), the Sun's ecliptic longitude in J2000 coordinates is computed and compared against IAU constellation boundaries. We validated this engine against Stellarium: at 10,950 BC, the summer solstice sun falls at ecliptic longitude 270.89°, within the Sagittarius boundary (266.5°–300.0°). A second check confirms the vernal equinox at 3,000 BC falls within Taurus (53.5°–90.4°), consistent with the historical "Age of Taurus."

3.2 Paleoclimatic event catalog

We compiled a catalog of 14 independent event clusters spanning 20,000–5,000 BC (Table 2). Events were drawn from ice core chronologies (GICC05), speleothem records, radiocarbon-dated geological events, and marine sediment cores. Events with distinct physical causes separated by fewer than 500 years were grouped into clusters to avoid double-counting. Each cluster carries a tolerance window reflecting its dating uncertainty.

3.3 Visual similarity model

We adopt Sweatman's rank-sum framework directly. Each of the six testable animals is independently ranked against 44 visible constellations by visual similarity. Sweatman's published ranks are $[1, 1, 1, 4, 5, 6]$, yielding a rank-sum of 18. The probability that independent uniform draws from $\{1, \ldots, 44\}$ produce a sum ≤ 18 is computed exactly via the inclusion-exclusion formula, avoiding Monte Carlo noise. We additionally verify via brute-force enumeration for the three-animal subgroups.

Table 2: Paleoclimatic event clusters. Tolerance half-widths reflect dating uncertainty.

Cluster	Date (BC)	Half-width (yr)	Key reference
LGM peak	19,050	1,000	Clark et al. (2009)
Bonneville Flood	15,550	500	O'Connor (1993)
Heinrich Event 1	14,250	500	Hemming (2004)
Bølling-Allerød + MWP-1A	12,700	200	Deschamps et al. (2012)
Older Dryas	12,075	200	Rasmussen et al. (2014)
YD cluster	10,900	250	Reinig et al. (2021)
YD termination	9,750	100	Cheng et al. (2020)
PBO / Bond 8 / MWP-1B	9,400	200	Rasmussen et al. (2007)
Bond 7	8,350	500	Bond et al. (1997)
Mammoth extinction	7,700	300	Kuzmin & Orlova (2004)
Bond 6	7,450	500	Bond et al. (1997)
8.2 ka cluster	6,300	300	Barber et al. (1999)
Mazama VEI 7	5,780	200	Zdanowicz et al. (1999)
Late cluster	5,500	400	Ryan & Pitman (1997)

3.4 Null models

We test four null models of increasing breadth.

The **anchor null** evaluates the date component. A random "sky reader" assigns the sun disc to one of 13 ecliptic constellations and one of 4 cardinal days (52 combinations total), computes the precession date, and checks whether it falls within the tolerance window of any cataloged event. We report two variants: a strict model using only the constellation center date, and a broad model checking whether any date within the full constellation window overlaps any event window.

The **similarity null** evaluates the visual component. Each animal independently draws a rank from $\{1, \ldots, 44\}$ (uniform). The probability that the resulting rank-sum is ≤ 18 is the visual-similarity p-value.

The **joint null** combines both under an independence assumption: $P(\text{joint}) = P(\text{anchor match}) \times P(\text{rank-sum} \leq 18)$. We state explicitly that this assumes the date-determining step and the visual-ranking step are independent. If ranking was influenced by foreknowledge of the desired date, the true joint probability is higher (less significant).

The **multiverse** enumerates all defensible analyst choices—anchor model (center vs. window), tolerance (100, 200, 250, 300 yr), event catalog (all 14 clusters vs. major-only), cardinal day scope (summer solstice only vs. all four), and animal inclusion (all 6 vs. subsets)—and reports the distribution of joint probabilities across the resulting paths.

3.5 Software and reproducibility

All analyses are implemented in a single self-contained Python script (~1,500 lines) requiring only `numpy` and `matplotlib`. All data—constellation boundaries, event catalog, Sweatman's rankings—are embedded in the script. No external files, APIs, or proprietary software are needed. The script, run log, and complete numerical output (JSON) are available at `https://github.com/simossss/gobekli-pillar43-statistics`.

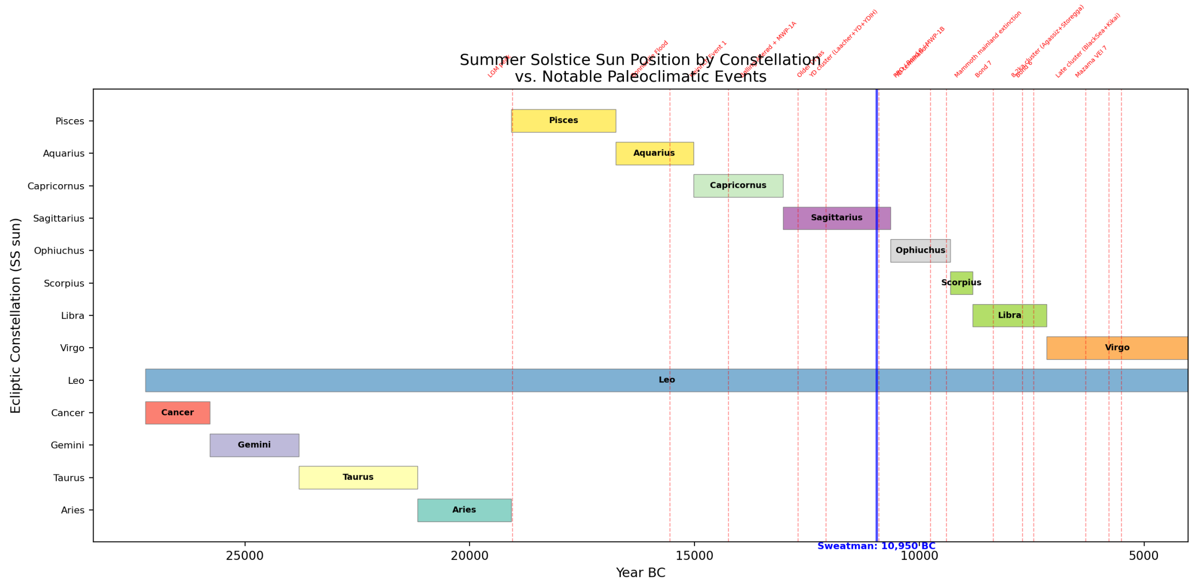

Figure 1: Summer solstice sun position by ecliptic constellation (horizontal bars) versus notable paleoclimatic events (vertical dashed lines). Sweatman's claimed date of 10,950 BC is marked in blue. The Sagittarius window spans approximately 10,635–13,034 BC.

4 Results

4.1 The date coincidence is not rare

Anchor test (Experiment 1). Of 52 constellation × cardinal-day combinations, 16 (30.8%) produce a center-date match at ±250 years under the strict model, and 35 (67.3%) under the broad model (Figure 4). At ±100 years, the strict model still gives 11.5%. The date component of Sweatman's claim is not a rare coincidence under either model. The 16 strict-model matches at ±250 years (Supplementary Material S1) span 12 of 13 ecliptic constellations and all four cardinal days, matching events from the Bonneville Flood (15,550 BC) to the Mazama eruption (5,780 BC). Figure 1 shows the temporal relationship between constellation windows and cataloged events.

Coverage fraction (Experiment 9). 42.5% of the 15,000-year window from 20,000 to 5,000 BC lies within ±250 years of at least one cataloged event. At ±500 years, coverage reaches 71.2%. The deglacial and early Holocene period is so densely packed with climatic and catastrophic events that a randomly chosen date from this era has a 42.5% chance of being "near something notable."

Leave-one-event-out (Experiment 7). No single event removal reduces the anchor probability below 25.0%. The most influential cluster to remove is the Older Dryas (which causes a 5.8 percentage-point drop). Notably, removing the YD cluster—the specific event Sweatman claims the pillar commemorates—does not change the baseline at all. This is because the Sagittarius center date (11,835 BC) is closer to the Older Dryas than to the YD onset.

Circular-shift test (Experiment 10). We rigidly shift the entire event catalog by a random offset within a fixed 15,000-year window, wrapping modulo the window length to preserve event count and inter-event spacing. Across 1,000 shifts, 23.9% produce an anchor probability at least as high as the baseline 30.8%. The real catalog's event

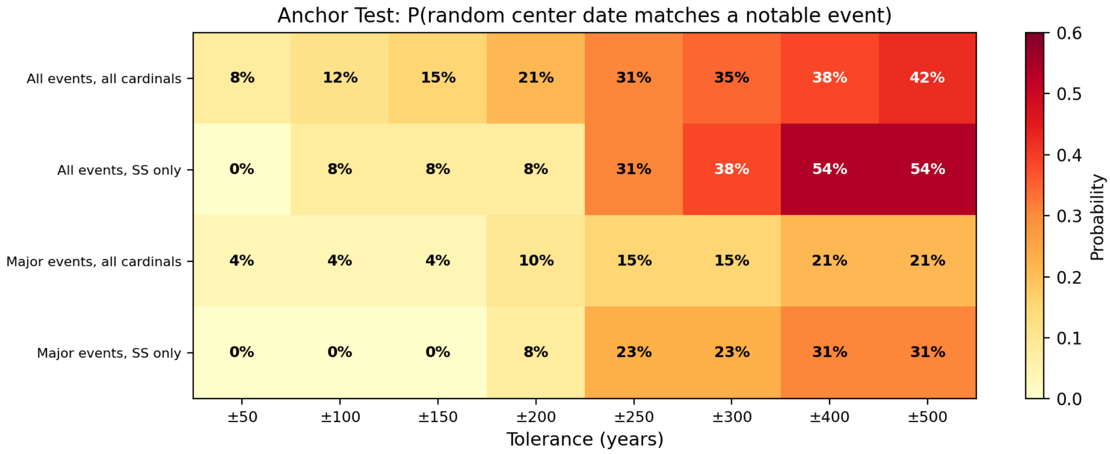

Figure 2: Sensitivity heatmap: center-only anchor probability as a function of tolerance window and scenario (event catalog $\times$ cardinal day scope). Even the most conservative scenario (major events only, summer solstice only) yields 23% at ± 250 years.

placement is broadly typical relative to circular shifts; event density, not the specific absolute positioning of events in time, does most of the work.

Catalog robustness (Experiment 5). Across 500 perturbed catalogs (random event drops and date shifts within uncertainty), the anchor probability ranges from 9.6% to 38.5% (mean 25.9%, Figure 5). Even under the most favorable catalog perturbation, the probability never drops below approximately 10%.

4.2 The visual similarity is genuine but concentrated

Exact rank-sum (Experiment 2). The probability that six independent uniform draws from $\{1, \ldots, 44\}$ produce a sum ≤ 18 is exactly 2.558×10^{-6}, approximately 1 in 391,000. This is the only numerically rare component of the claim. The expected rank-sum under the null is 135 ± 31; Sweatman's observed sum of 18 lies 3.76 standard deviations below the mean (Figure 3).

Decomposition (Experiment 6). We split the six mappings into the three strongest (ranks $[1, 1, 1]$, sum $= 3$) and three weakest (ranks $[4, 5, 6]$, sum $= 15$), as shown in Table 3. The top three contribute $P = 1.17 \times 10^{-5}$ (1 in 85,184). The bottom three contribute $P = 5.34 \times 10^{-3}$ (1 in 187). Although the bottom-three subset remains statistically significant under conventional thresholds, its evidential weight is far smaller than that of the top three. The combined significance is concentrated disproportionately in three mappings—scorpion $\rightarrow$ Scorpius, canid $\rightarrow$ Lupus, and tall bird with snake $\rightarrow$ Ophiuchus—rather than being evenly distributed across all six.

Table 3: Rank-sum decomposition. Exact probabilities computed via inclusion-exclusion.

Group	Ranks	Sum	$P(\text{sum} \leq S)$	$\approx$ 1 in
Top 3 (scorpion, canid, Ophiuchus)	$[1, 1, 1]$	3	1.17×10^{-5}	85,184
Bottom 3 (Pisces, Gemini, Virgo)	$[4, 5, 6]$	15	5.34×10^{-3}	187
All 6	$[1, 1, 1, 4, 5, 6]$	18	2.56×10^{-6}	390,881

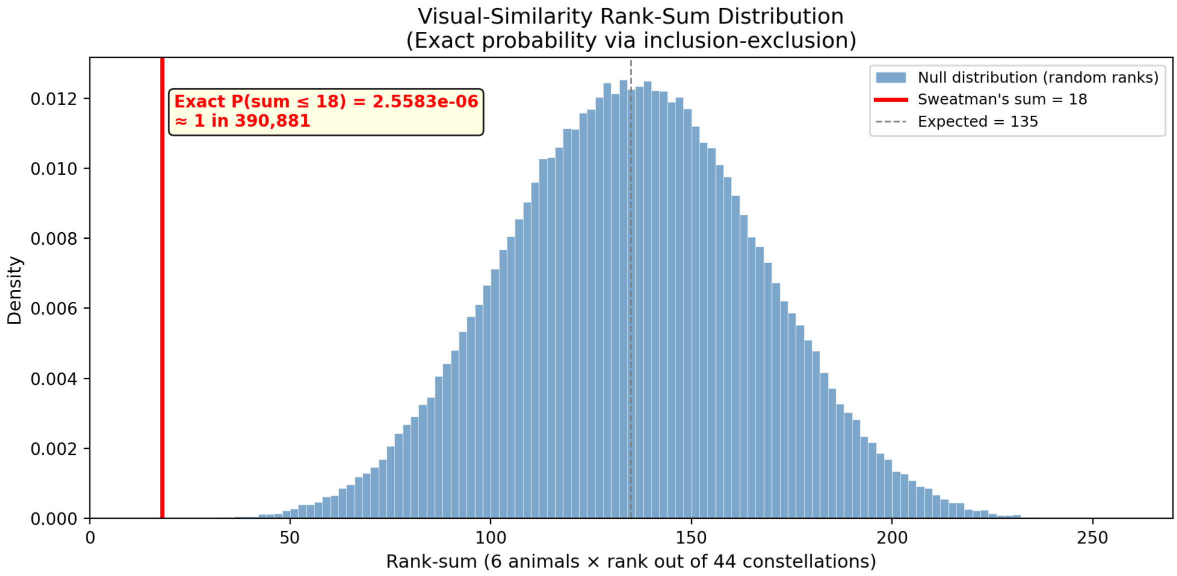

Figure 3: Null distribution of the rank-sum statistic (6 independent uniform draws from $\{1, \ldots, 44\}$). Sweatman's observed sum of 18 (red line) lies far in the left tail, with exact $P = 2.56 \times 10^{-6}$.

Leave-one-animal-out (Experiment 11). Each rank-1 animal is equally critical: removing any one of the three weakens the result by a factor of 14.7 (Table 4). Removing the bending bird (rank 4) weakens it by 4.7×, the ibex (rank 5) by 3.1×, and the frog/bear (rank 6) by only 1.9×. The influence gradient is monotonic with rank and confirms that the three rank-1 animals carry nearly all the statistical weight.

Table 4: Leave-one-animal-out influence analysis.

Dropped animal	Rank	Remaining P	$\approx$ 1 in	Weakening
Scorpion	1	3.75×10^{-5}	26,651	14.7×
Wolf/dog	1	3.75×10^{-5}	26,651	14.7×
Tall bird + snake	1	3.75×10^{-5}	26,651	14.7×
Bending bird	4	1.21×10^{-5}	82,376	4.7×
Ibex/quadruped	5	7.80×10^{-6}	128,140	3.1×
Frog/bear	6	4.80×10^{-6}	208,228	1.9×

Identity permutation (Experiment 13). Conditional on a rank pattern containing three rank-1 slots, the probability that the specific claimed trio occupies all three is $1/\binom{6}{3} = 1/20 = 5\%$. This is mildly informative but not dramatic, and is best understood as a complement to the decomposition rather than an independent line of evidence.

4.3 Significance is path-dependent

Joint probability (Experiment 3). Under the strict anchor model at ±250 years, the joint probability is 7.87×10^{-7} (approximately 1 in 1.27 million). Under the broad model, it is 1.72×10^{-6} (1 in 581,000). Both are weaker than Sweatman's reported

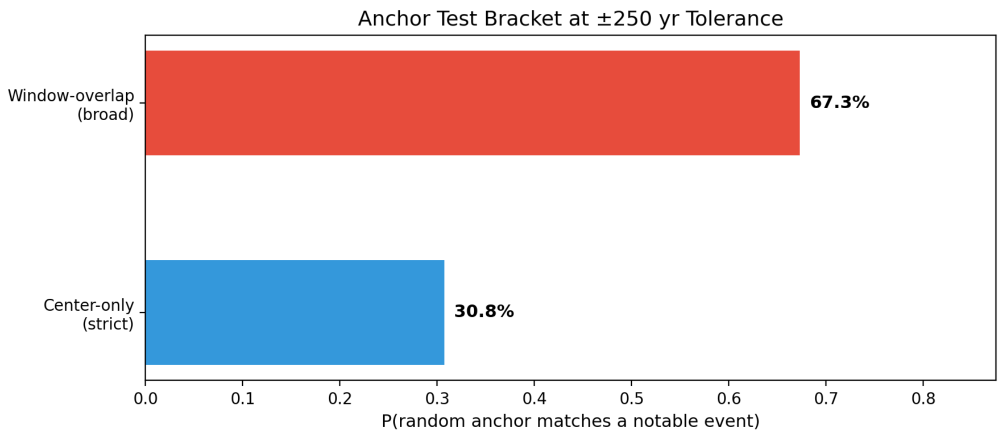

Figure 4: Anchor test bracket at ± 250 years: center-only (strict, 30.8%) and window-overlap (broad, 67.3%) models.

1 in 1.4 million because his calculation does not account for the look-elsewhere effect on the date component.

Sensitivity (Experiment 4). The anchor probability varies from 0% (major events only, summer solstice only, ± 50 yr) to 54% (all events, summer solstice only, ± 500 yr), as shown in Figure 2. Even the most conservative scenario tested—major events only with summer solstice only—gives 23% at ± 250 years.

Multiverse (Experiment 8). We enumerate 128 analyst-choice paths formed by crossing anchor model (2), tolerance (4), event catalog (2), cardinal day scope (2), and animal inclusion (4 subsets). Within the 32 primary paths (all 6 animals, varying only methodological choices), the joint probability ranges from 9.84×10^{-8} to 1.97×10^{-6}—a 20-fold range. 42% of primary paths are as or more significant than Sweatman's (Table 5). In the full 128-path multiverse (which includes the stronger degree of freedom of dropping animals post hoc), the range expands to 9.84×10^{-8} to 9.03×10^{-6}—two full orders of magnitude. Not all paths are equally defensible: dropping animals post hoc is a much stronger degree of freedom than choosing center vs. window anchor. But even within the primary multiverse, the joint probability is path-dependent rather than singular—it is a distribution that depends on analyst choices.

Table 5: Primary multiverse summary (32 paths, all 6 animals). Full path table in Supplementary Material S3.

Statistic	Joint P	$\approx$ 1 in
Minimum (most significant)	9.84×10^{-8}	10,162,905
Median	9.35×10^{-7}	1,069,519
Sweatman's path	7.87×10^{-7}	1,270,363
Maximum (least significant)	1.97×10^{-6}	507,614

Ranking sensitivity (Experiment 12). Under conservative rank adjustments (+1 to each published rank), the overall significance drops from 1 in 391,000 to 1 in 54,000, a sevenfold reduction from a minimal perturbation (Table 6). Under skeptical rankings (top three at rank 3 instead of rank 1), the same overall probability obtains but the decomposition shifts: the top-three contribution weakens from 1 in 85,184 to 1 in 1,015. These

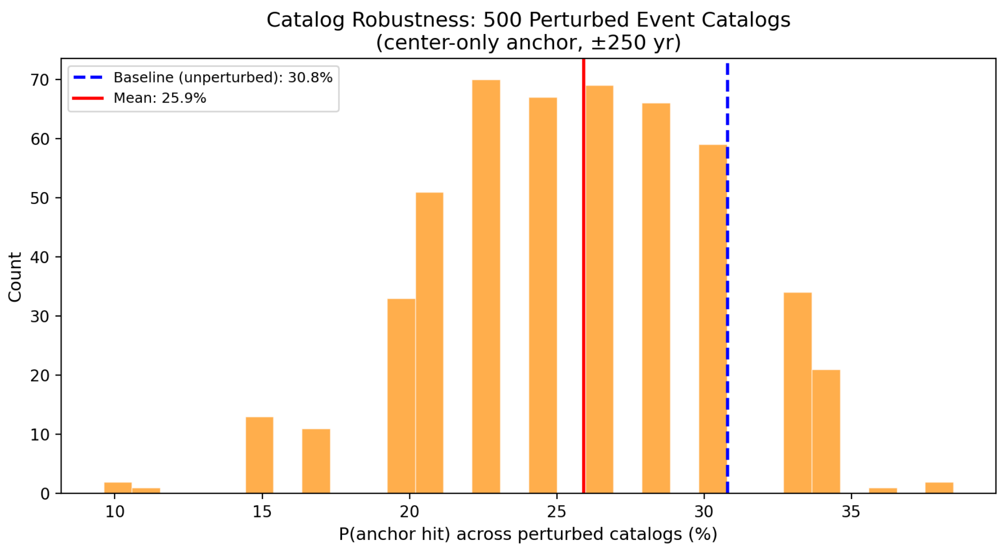

Figure 5: Distribution of anchor probability across 500 perturbed event catalogs (center-only, ±250 yr). The baseline (unperturbed) catalog yields 30.8%. The mean across perturbations is 25.9%.

are hand-crafted stress tests rather than independent validations, but they demonstrate that the published significance is sensitive to modest changes in the ranking vector.

Table 6: Ranking sensitivity across alternative schemes.

Scheme	Ranks	Sum	$P(\text{sum} \leq S)$	$\approx$ 1 in
Sweatman original	$[1, 1, 1, 4, 5, 6]$	18	2.56×10^{-6}	390,881
Conservative (+1 each)	$[2, 2, 2, 5, 6, 7]$	24	1.85×10^{-5}	53,912
Skeptical (top 3 at rank 3)	$[3, 3, 3, 4, 5, 6]$	24	1.85×10^{-5}	53,912
Mixed (top 3 vary)	$[1, 2, 3, 4, 5, 6]$	21	7.48×10^{-6}	133,722
Degraded (top 3 at rank 5)	$[5, 5, 5, 4, 5, 6]$	30	8.18×10^{-5}	12,221
Generous (bottom 3 improved)	$[1, 1, 1, 2, 3, 4]$	12	1.27×10^{-7}	7,853,154

5 Discussion

5.1 What the evidence actually supports

Within the published scoring framework, three animal-constellation visual matches rank 1st out of 44 candidates each. The scorpion → Scorpius, canid → Lupus, and bird-with-snake → Ophiuchus pairings are difficult to dismiss as pure coincidence. The probability that three independent uniform draws from 44 options all yield rank 1 is 1 in 85,184. This is the core of Sweatman's case, and it is real.

The rest of the evidence is weaker than it appears. The date component contributes relatively little once the density of paleoclimatic events in the deglacial period is accounted for. Three of the six animal-constellation mappings contribute substantially less statistical weight than the other three. The overall 1-in-1.4-million headline is conditional on specific analyst choices and drops by orders of magnitude under alternative defensible assumptions.

5.2 The self-scoring problem

The most important limitation of the claim is that Sweatman and colleagues ranked all 264 animal-constellation similarity pairs (6×44) themselves. The published papers provide verbal justifications for why each animal matches its proposed constellation, but these judgments were made by researchers who knew the hypothesis and its desired outcome.

This is not an accusation of dishonesty. Confirmation bias in visual pattern-matching is a well-documented phenomenon, and the archaeoastronomy literature has long recognized that subjective alignment judgments are vulnerable to it (Ruggles, 2015; Silva, 2020). The problem is structural: the same people who proposed the hypothesis also generated the data (similarity rankings) that the statistical test evaluates. This conflation of hypothesis generation and hypothesis testing inflates apparent significance in ways that are difficult to quantify without external validation.

A deeper problem is that the published similarity rankings operate across at least three interpretive layers: whether ancient observers identified constellations at all, whether they grouped the same stars into the same figures later used in the Greek tradition, and whether they assigned the same animals to those figures. Our analysis addresses only the final layer within the modern Western constellation framework adopted by Sweatman. This creates a semantic-geometric circularity: some proposed matches may appear strong partly because the modern constellation already carries the same animal label (scorpion $\rightarrow$ Scorpius, wolf $\rightarrow$ Lupus), not because the underlying star pattern independently compels that reading. A blind ranking experiment would therefore test the reproducibility of the published similarity judgments within that adopted framework, but it could not by itself establish that Pre-Pottery Neolithic observers partitioned and named the sky in the same way as much later Mediterranean traditions.

5.3 Implications for archaeoastronomy methodology

Several methodological lessons emerge from this analysis that apply beyond the Pillar 43 case.

Date coincidences in the deglacial period are inexpensive. Any study claiming that an archaeoastronomical date aligns with a notable event should report the coverage fraction—what share of the relevant time window lies near some event—as a baseline. Without this baseline, event-dense periods will generate impressive-looking coincidences routinely.

Visual similarity scores should be generated by blinded raters, not by hypothesis proponents. The probabilistic framework used by Sweatman is methodologically sound in principle. The weakness is in the input data, not the statistical machinery. Silva's (2020) probabilistic framework for skyscape archaeology addresses alignment orientations but does not cover the iconographic similarity problem that is central to the Pillar 43 case.

Analyst degrees of freedom should be reported via multiverse analysis, not hidden behind a single p-value. The two-order-of-magnitude range we find across defensible paths is not unusual in archaeological statistics, but it is rarely disclosed. Reporting only the most favorable path, as is conventional, creates an inflated impression of evidential strength.

Leave-one-out diagnostics on both the event side and the mapping side should be standard practice. They reveal which components drive the result and whether the conclusion is robust to the removal of any single element.

5.4 Limitations

We accept Sweatman's published rankings at face value throughout the core analysis. Our ranking-sensitivity experiments (Experiment 12) explore perturbations, but these are hand-crafted stress tests rather than independent assessments. The event catalog is our construction and could be contested: a different catalog would yield different anchor probabilities, though our catalog robustness sweep (Experiment 5) and circular-shift test (Experiment 10) demonstrate that the qualitative conclusion is stable across a wide range of perturbations.

The linear precession approximation introduces approximately 1° of error over 13,000 years. This is negligible relative to the $\sim 30^\circ$ width of most ecliptic constellations but means our dates should be treated as approximate to within $\sim$70 years.

We do not evaluate the 2024 calendar interpretation (Sweatman, 2024), which proposes that geometric symbols on Pillar 43 encode a 365-day lunisolar counting system. If correct, this would constitute an independent line of evidence for the astronomical interpretation that does not depend on visual similarity rankings. It is beyond the scope of the present analysis.

We test the statistical framework, not the archaeological plausibility of the astronomical interpretation itself. Questions about whether Neolithic communities could have recognized constellations similar to those of the ancient Greeks, whether the pillars are in original positions, or whether the radiocarbon chronology is compatible with the proposed date are addressed by Notroff et al. (2017) and others. These archaeological arguments are orthogonal to the statistical questions we address here.

5.5 The blind ranking experiment as the decisive next test

The central unresolved question is whether the three strong visual matches are objectively compelling or subjectively imposed. A blind ranking experiment would answer this directly.

The protocol would be straightforward. Standardized visual descriptions or images of the six Göbekli Tepe animal carvings and the 44 visible constellation figures would be presented to raters with no knowledge of Göbekli Tepe, archaeoastronomy, or the Sweatman hypothesis. Each rater would rank, for each animal, which constellation figures it most resembles. The scoring rule would be preregistered. The key outcome is whether scorpion, canid, and bird-with-snake still rank 1st for Scorpius, Lupus, and Ophiuchus respectively—or whether those rankings shift substantially when the hypothesis context is removed.

If the same three animals replicate their rank-1 positions under blind conditions, the visual signal is objective and the statistical case becomes considerably stronger. If they do not, the claim's statistical foundation depends on judgments that are inseparable from knowledge of the desired outcome, and the reported p-value cannot be taken at face value.

This experiment is feasible with modest resources, requires no fieldwork or proprietary data, and would settle the ranking-reproducibility question. It would not, however, resolve the deeper historical question of whether Pre-Pottery Neolithic observers shared the constellational groupings and animal assignments of much later Mediterranean traditions.

6 Conclusion

Sweatman's Pillar 43 interpretation contains a genuine statistical signal: within the published scoring framework, three animal-constellation visual matches rank 1st out of 44 candidates each. This is unlikely to arise by chance ($P \approx 1$ in 85,000) and represents the strongest component of the claim.

The reported overall significance of 1 in 1.4 million substantially overstates the evidence, for three reasons. First, the date component is not rare once the look-elsewhere effect is accounted for: 43% of the deglacial era lies within ±250 years of a notable event, and 31% of random anchor assignments produce a date match under the strict model. Second, the statistical weight is concentrated disproportionately in three of the six visual mappings; the bottom three yield only 1 in 187. Third, the overall significance varies by two orders of magnitude across defensible analyst choices, from approximately 1 in 10 million to 1 in 111,000 depending on anchor model, tolerance, event catalog, cardinal day scope, and animal inclusion.

The critical unresolved question is whether the three strong visual matches are objectively compelling or subjectively imposed. Until a blind ranking experiment is performed, the Pillar 43 claim remains an intriguing but unvalidated hypothesis whose statistical support is weaker and more conditional than reported.

AI Disclosure

This research was conducted using Claude (Anthropic) as a technical reasoning partner for statistical design, code development, and manuscript drafting. ChatGPT (OpenAI) served as an independent reviewer of the statistical methodology and code across multiple review cycles. The analysis was executed on an NVIDIA DGX Spark. The author takes full responsibility for the scientific content and conclusions.

Data Availability

The analysis script, run log, results JSON, and all figures are available at: `https://github.com/simossss/gobekli-pillar43-statistics`

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Supplementary Materials

- **S1:** Full run log (console output of all 13 experiments)
- **S2:** results.json (complete numerical output including full multiverse path table)
- **S3:** Primary multiverse path table (32 paths with joint probabilities)
- **S4:** Leave-one-event-out complete table
- **S5:** Leave-one-animal-out complete table
- **S6:** Ranking sensitivity table (6 schemes)